

DATA SCIENCE RESOURCES

Transportation Projects

Project 1. Cab Fare Prediction Using NYC Taxi Dataset

Problem Statement: Predict taxi fares accurately based on trip distance, time, and location to help riders and companies estimate costs.

Objective: Build a regression model that forecasts fare amounts from features like distance, passenger count, and timestamps.

Dataset Source: [NYC Taxi Trip Dataset on Kaggle](#).

Techniques and Concepts

- Data cleaning and outlier removal
- Feature engineering: haversine distance, time features
- Linear regression and XGBoost models
- Model evaluation with RMSE and MAE

Tools and Libraries: Python, Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Streamlit.

Expected Outcome: A deployed Streamlit app that takes trip inputs and predicts fare estimates in real-time.

Project 2. Accident Hotspot Mapping Using Geospatial Clustering

Problem Statement: Identify high-risk road sections by mapping accident hotspots using clustering algorithms.

Objective: Cluster accident locations and visualize hotspots on interactive maps.

Dataset Source: [Government road accident datasets](#) (local transport authority or Kaggle road accidents data).

Techniques and Concepts

- Geospatial data handling
- Clustering algorithms (DBSCAN)
- Interactive mapping with Folium or Kepler.gl

Tools and Libraries: Python, GeoPandas, Folium, Scikit-learn, Matplotlib.

Expected Outcome: An interactive heatmap or cluster map showing accident hotspots for urban planning or safety improvements.

Project 3. Route Optimization Model for Delivery Fleets

Problem Statement: Find the shortest or fastest route for delivery vehicles to minimize travel time and costs.

Objective: Implement routing algorithms to optimize delivery routes based on location and traffic data.

Dataset Source: Sample delivery points or use [OpenStreetMap data via APIs](#).

Techniques and Concepts

- Graph theory algorithms (Dijkstra, A*)
- Vehicle Routing Problem (VRP) using Google OR-Tools
- Real-time traffic data integration (optional)

Tools and Libraries: Python, NetworkX, Google OR-Tools, Matplotlib.

Expected Outcome: A routing model that outputs optimized delivery paths, which can be visualized on a map or used in logistics planning.
